1. Clinical Question & Core Hypothesis

Clinical Question

If EMT and Wnt/β-catenin activation drive metastatic spread, why do EMT-high bladder tumors also frequently resist radiotherapy, and which measurable histomorphologic feature on H&E captures the EMT/Wnt axis that predicts RT failure? 2) Tier 1 paths used: I built on (a) EMT → metastatic spread, (b) Wnt/β-catenin signaling → EMT/metastasis, and (c) EMT → altered radiation therapy response — these three together form a mechanistic chain linking Wnt activation to EMT phenotypes that both increase dissemination and confer radioresistance. 3) Tier 2 support: BCG–urothelial binding, BCG as bladder cancer therapy, and the clinical context of combining treatments provide an immune/therapeutic backdrop (e.g., immune activation or suppression can modulate EMT effects and influence residual disease after RT), justifying downstream testing of combination approaches but not as the primary mechanistic axis. 4) Why this endpoint: I selected therapy_response (radiotherapy response) because the strongest mechanistic link in Tier 1 directly ties EMT/Wnt status to RT sensitivity; therapy_response is the most specific, actionable clinical endpoint to test the causal model. 5) Deliberate exclusions: I excluded making BCG or broader immunotherapy part of the core mechanistic claim to keep the hypothesis sharply focused on Wnt–EMT–RT causality; immunotherapy/BCG and broad metastasis pathways are placed as secondary modifiers or follow-up studies to avoid diluting the core question?

Working Hypothesis

In bladder tumors, a high stromal-to-tumor ratio (TSR) captures an EMT/Wnt/β-catenin–driven program evidenced by CTNNB1 activation and CDH1 suppression, which reprograms cancer cells toward a mesenchymal phenotype with an enhanced DNA damage response, thereby conferring intrinsic radiotherapy resistance. Consequently, TSR-high tumors with elevated CTNNB1 and reduced CDH1 will exhibit poorer radiotherapy response (therapy_response) and shorter time-to-progression following RT compared with TSR-low, CTNNB1-low/CDH1-high tumors, indicating that targeting the Wnt/EMT axis could restore RT sensitivity.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	CTNNB1, CDH1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CTNNB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

CDH1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	SNAI1	Supporting modifier from knowledge graph
Secondary	ZEB1	Supporting modifier from knowledge graph
Secondary	VIM	Supporting modifier from knowledge graph
Secondary	MMP2	Supporting modifier from knowledge graph
Secondary	MMP9	Supporting modifier from knowledge graph
Secondary	LRP5	Supporting modifier from knowledge graph
Secondary	FZD7	Supporting modifier from knowledge graph
Secondary	TLR4	Supporting modifier from knowledge graph
Secondary	CD274	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective analyses in cohorts of bladder cancer patients treated with RT to correlate TSR (pathology-based) and CTNNB1/CDH1 expression with therapy_response;

Prediction 2: In vitro studies using bladder cancer cell lines with manipulated CTNNB1 activation and CDH1 expression to measure clonogenic survival after irradiation and assess EMT markers and DNA damage response;

Prediction 3: Patient-derived organoids or xenografts with RT ± Wnt/EMT inhibitors to test rescue of RT sensitivity;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SNAI1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ZEB1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.